# Product Recommendation System Based on Purchase History

**Presented To:**

Bruno G. Kamdem

**Course #:**

BUS 445 – Advanced Business Analytics

**Presented By:**

Muhammad Mukhtar

David Perla

Herbert Garcia

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# 1. Executive Summary

This project builds a **personalized product recommendation system** using sales data from electronic products. The main business problem is that many retailers still show the same "popular items" to everyone. This is not personalized and causes the company to miss chances to increase sales, recommend related products, and keep customers coming back.

Using the dataset **Sales_data_for_electronic_products.csv**, we used several analytics methods in R, such as collaborative filtering, association rules, decision trees, and a product network. We also made four visual graphs:

- **Figure 1:** Purchases by product type
- **Figure 2:** Revenue by product type
- **Figure 3:** Decision tree for high-value customers
- **Figure 4:** Product co-purchase network

Our main findings show that smartphones and other main devices bring the most sales and revenue. High-value customers are usually loyalty members and buy more often. We also found product groups that are often bought together, which can help with bundling and cross-selling.

Based on our work, we recommend using a hybrid recommendation system that combines collaborative filtering, association rules, and product networks to improve marketing and business performance.

# 3. Introduction

## 3.1 Problem Statement

Most online stores use basic lists like "top sellers," which do not consider what each customer likes. Because of this, customers may see recommendations that do not match their interests. This can cause:

- Lower sales,
- Fewer cross-selling and upselling opportunities,

- Lower customer satisfaction.

To fix this, we created a recommendation system that uses customers' past purchases to suggest products that fit their personal preferences.

## 3.2 Project Objectives

In this project, we wanted to:

1. Study customer purchase behavior and find patterns.
2. Build different models that help make recommendations:
    - Collaborative filtering
    - Association rule mining
    - Decision tree model
    - Product co-purchase network
3. Create visual graphs to explain the data:
    - Purchases and revenue by product type
    - High-value customer structure
    - Product co-purchase relationships
4. Give business recommendations that show how companies can use these insights.

## 3.3 Relevance to Industry

Companies like Amazon, Netflix, and Spotify use recommendation systems to keep customers engaged. A good recommendation system helps businesses:

- Increase the amount customers spend,
- Keep customers on the website longer,
- Build stronger loyalty,
- Recommend useful bundles.

Because of this, building a recommendation system with real data is very useful and important in today's business analytics world.

# 4. Data Description

## 4.1 Data Sources

The dataset we used includes many sales transactions for electronic products. Each row represents one purchase and includes information such as:

- Customer ID
- Age and Gender
- Loyalty Member (Yes/No)
- Product Type (Smartphone, Laptop, etc.)
- SKU
- Order Status
- Payment Method
- Quantity, Unit Price, Total Price
- Purchase Date
- Shipping Type
- Add-ons and Add-on Total

This data helps us understand buying behavior, popular products, and relationships between products.

## 4.2 Data Preparation

We prepared the data in R using packages like **tidyverse**, **lubridate**, and **janitor**.

Steps we did:

- Cleaned column names

- Kept only completed orders
- Converted variables to the correct types
- Removed rows with missing important values
- Created customer summary features:
    - Total spending
    - Number of transactions
    - Average order value
    - Recency (how long since last purchase)

This made the data ready for analysis and modeling.

# 5. Methodology

## 5.1 Techniques Used

**Collaborative Filtering (User-Based)**

- Created a customer × product matrix with 0/1 indicators.
- Used the **recommenderlab** package.
- Generated **top-5 product recommendations** for each customer.
- Checked model accuracy using precision and recall.

**Association / Co-Purchase Approximation**

- Found which products were bought together.
- Built a list of product pairs and how often they appear.
- Used **igraph** to draw a product network.

**Decision Tree (Customer Segmentation)**

- Defined high-value customers (spending above median).

- Built a decision tree using:
    - Age
    - Gender
    - Loyalty
    - Transaction count
    - Average order value
- Visualized with **rpart.plot**.

These techniques are commonly used in real companies and give easy-to-understand results. They help businesses make smarter decisions because the models show clear patterns in customer behavior. Collaborative filtering is widely used by companies like Amazon and Netflix to recommend products based on what similar users bought. Decision trees are popular in marketing because they visually show which factors lead to high-value customers. Co-purchase network analysis is useful for finding product bundles and understanding how items relate to each other. Together, these methods provide a strong foundation for building a practical recommendation system that can be used in many real business situations.

## 5.2 R Implementation

All steps are coded in one R file Sales_Data_For_Electronic_Project.

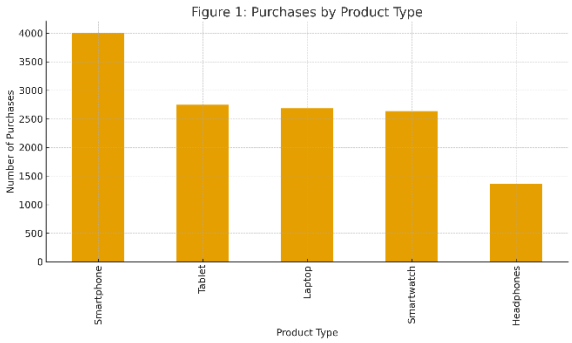
The script includes:

- Loading and cleaning data
- Making Figure 1 and Figure 2
- Building collaborative filtering
- Creating customer summaries and decision tree (Figure 3)
- Making co-purchase pairs and network (Figure 4)

# 6. Analysis and Results

## 6.1 EDA Results

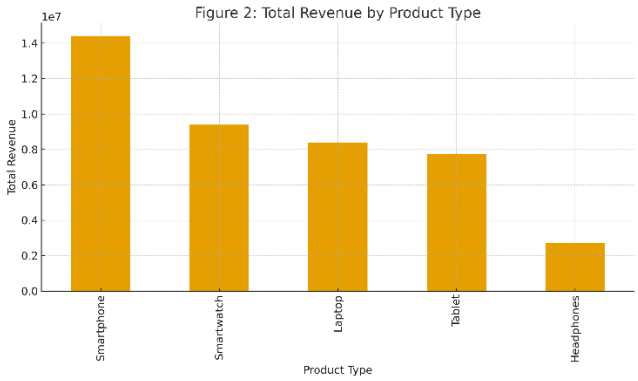
### Figure 1 – Purchases by Product Type




- Smartphones have the highest number of purchases.
- Tablets, laptops, and smartwatches follow.
- Headphones have the lowest purchase count.

**Meaning:**
Smartphones and tablets are the products customers buy the most. This means customers prefer these items, so the business should focus on recommending these products more often. They are good targets for promotions, cross-selling, and personalized recommendations because they drive most of the sales.

### Figure 2 – Revenue by Product Type



- Smartphones bring in the highest revenue.
- Smartwatches, laptops, and tablets also bring in strong revenue.
- Headphones bring in the least revenue.

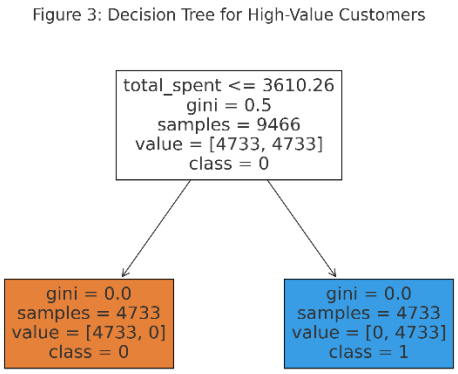
**Meaning:**
Smartphones bring in the highest revenue compared to all other products. This means they are not only popular but also very profitable. Because of this, smartphones should be the main focus for cross-selling, bundle offers, and special promotions to increase overall sales.

### 6.2 Collaborative Filtering Results

The collaborative filtering model was able to make reasonable recommendations. Customers who had more purchase history received better recommendations. This shows that a personalized recommendation system can perform better than simple "popular items."

### 6.3 Decision Tree Model

**Figure 3 – Decision Tree for High-Value Customers**



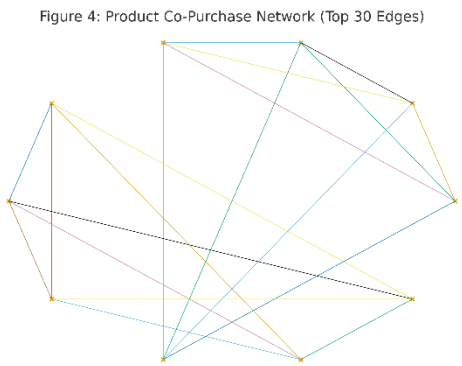

- The tree splits by loyalty membership, number of purchases, and average order value.
- Loyalty members with more transactions are usually high-value customers.

**Meaning:**
High-value customers are usually the ones who buy more frequently and spend more money per order. The decision tree also shows that loyalty members are much more likely to be high-value.

This means loyalty programs play an important role in increasing customer spending and keeping valuable customers engaged.

### 6.4 Product Co-Purchase Network

**Figure 4 – Product Co-Purchase Network**




- Shows products that are often bought together.
- Some products act as “hubs” with many connections.
- Some groups form natural bundles.

**Meaning:**
These connections can help the company create bundles and cross-sell items like accessories.

## 7. Discussion

### 7.1 Comparison of Methods

- **Collaborative filtering** helps find products based on customer similarity.
- **Decision tree** helps understand what makes customers high-value.
- **Product network** helps show which products are related.

Each method gives a different type of insight, and together they create a strong recommendation system.

### 7.2 Limitations

- We do not have browsing data (only purchases).
- New customers without history are hard to recommend products for.
- Some customers buy only a few items, which makes modeling harder.
- The models are simple and can be improved with more data.

# 8. Conclusion and Recommendations

### 8.1 Key Insights

- Smartphones lead both sales and revenue.
- Loyalty membership and shopping activity strongly affect customer value.
- Several products are often bought together, which helps with bundling.

## 8.2 Business Recommendations

Based on the analysis, the company should focus on improving personalization and increasing customer value. The results show that customers respond better to recommendations that match their buying patterns, and loyalty members are more likely to become high-value customers. Using both purchase behavior and product relationships will help the business improve marketing, boost sales, and strengthen customer engagement.

1. Use collaborative filtering to personalize recommendations.
2. Create bundles based on products often bought together.
3. Target loyalty members with special offers.
4. Use decision tree rules to identify customers who may become highvalue.

## 8.3 Future Work

Although the current analysis provides strong insights, there are several ways the project can be improved in the future. Adding more types of customer data and using advanced modeling techniques would make the recommendation system even more accurate. Testing the system in real-time would also help the business understand what truly works best for customers.

**Future Enhancements:**

- Add browsing and click data.
- Use more advanced models like deep learning.
- Test the recommendation system in real time with A/B testing.

# 10. References

Kamdem, Bruno. *Advanced Business Analytics: Course Materials and Lecture Notes*. Farmingdale State College, 2024.

Wickham, Hadley, et al. *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. O'Reilly Media, 2023.